

Modelling of tsunami-induced bore and structure interaction

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A series of three-dimensional smoothed particle hydrodynamics (SPH) and finite-element (FE) models, with a domain in the form of a water tank, were undertaken to simulate tsunami-induced bore impact on a discrete onshore structure on a dry bed. The fluid motion was simulated using the SPH-based software DualSPHysics. The tsunami-like waves were represented by solitary waves with different characteristics generated by the numerical paddle wavemaker. Numerical probes were uniformly distributed on the structure's vertical surface providing detailed measures of the pressure distribution across the structure. The peak impact locations on the structure's surface were specifically determined and the associated peak pressures then compared with the prediction of existing commonly used design equations. Using the pressure–time histories from the SPH model, FE analysis was conducted with Abaqus to model the dynamic response of a representative timber structure. The results show that the equations used to estimate the associated pressure for design purposes can be highly non-conservative. By gaining a detailed insight into the impact pressures and structure response, engineers have the potential means to optimise the design of structures under tsunami impact loads and improve survivability.

Notation

A_i	tributary area where pressure probe i is situated (m^2)
a	particle a
B	breadth of structure (m)
b	particle b
D	certain length of 1 m in the SPH model
D_F	distance of structure from shoreline (m)
d_p	water particle size or the initial spacing of particle (m)
E	Young's modulus of elasticity (m kg/s^2)
F_i	force at a certain point in a three-dimensional (3D) model (kg m/s^2)
g	gravitational acceleration (m/s^2)
H	solitary wave height (m)
h	characteristic smoothing length (m)
h_o	offshore water depth/still water level (m)
h_b	height of the bore (m)
h_{im}	maximum inundation depth (m)

L	shortest dimension of box in convergence study (m)
L_p	layers of pressure probes
m	mass of SPH particle (kg)
P_i	pressure at a certain area in a 3D model ($(\text{kg/m})/\text{s}^2$)
r_{ab}	distance between two particles (m)
ν	Poisson's ratio
v_j	velocity of bore (m/s)
W	weighting function or kernel
Π_{ab}	viscosity term
ρ	density of SPH particle (kg/m^3)
Γ	dissipative term

1. Introduction

In many parts of the world, structures in low-lying coastal areas are at risk of tsunami inundation and can suffer significant damage if they are not adequately designed for the

associated hydrodynamic loads. Onshore tsunami waves propagate from the deep ocean towards the shoreline and break when the incident wave height is approximately similar with the water depth. A broken tsunami wave moves inland in the form of a turbulent hydraulic bore (Palermo *et al.*, 2009), as captured on video footage of the 26 December 2004 Indian Ocean Tsunami (Nouri *et al.*, 2010). In the context of this work, a bore is defined as a steep, rapidly moving broken wave with an onshore mass flux that can be very destructive (IOC, 2013).

In the major tsunami disasters in 2004 and 2011, the tsunami bore inundation greatly affected shore-based structures, including many critical pieces of infrastructure such as railways, nuclear facilities and so on. By far, the majority of structures damaged or destroyed in these events were residential buildings. After the 2011 East Japan tsunami, Suppasri *et al.* (2013) stated that 115 163 houses were heavily damaged, 162 015 houses were moderately damaged and 559 321 houses were partially damaged. Many of those houses were constructed from timber. Such timber structures might survive the earthquake that generated the tsunami due to their relatively high strength-to-mass ratio and flexibility, but could not resist the hydrodynamic forces resulting from the bore impact.

An improved understanding of residential buildings' response to tsunami loads could help reduce casualties, increase the survivability of such structures and speed up recovery time post tsunami. Becker *et al.* (2011) show that a key factor of building response to the external actions is the interaction between the structural components of the building and the wave. The pressure of the tsunami bore can be likened to the pressures attributable to winds that are treated as uniform lateral pressures acting along the entire height of the building. However, in determining tsunami loads, the varying depth of the water, associated velocity and duration of impact are added variables that affect the resulting force and structure response.

Existing design codes and guides around the world present various approaches to quantification of tsunami loads on buildings. The City and County of Honolulu Building Code (CCH, 2000) recommends formulation as shown by Equation 1, adopted from Dames and Moore (1980), to predict hydraulic bore-like wave impact on vertical walls. *The Structural Design Method of Buildings for Tsunami Resistance (SMBTR)* in Japan developed by Okada *et al.* (2005) also assumed surge force as given by Equation 1. The surge force (F_s) given in Equation 1 results from the summation of hydrostatic and hydrodynamic force components as given by the following expression

$$1. \quad F_s = \frac{1}{2} \rho g h^2 b + \frac{1}{2} C_d \rho u^2 h b$$

where F_s is the surge force per unit width of wall, ρ is the density of water, g is the gravitational acceleration, h is the surge height usually assumed as equal to the inundation depth or flood level, b is the breadth of impacted structure, C_d is the drag coefficient recommended by CCH (2000) and FEMA 55 (FEMA, 2003) as being equal to 2.0 in the case of square or rectangular piles and u is the tsunami flow velocity. The general form of the tsunami-induced flow velocity is $u = C\sqrt{gh}$, where C is a constant coefficient that is assumed by Camfield (1980) and FEMA 55 (FEMA, 2003) as having a value of 2. The substitution of $u = 2\sqrt{gh}$ into Equation 1 results in the surge force (F_s) being estimated as follows

$$2. \quad F_s = \frac{1}{2} \rho g h^2 b + 4 \rho g h^2 b = 4.5 \rho g h^2 b$$

Fujima *et al.* (2009) also proposed two equations for a dry flat shoreline configuration based on the maximum water inundation level and the distance of the structure from the shoreline. The equation expresses the distance of a structure from the shoreline (D_F) in terms of (h_{im}/D_F), where h_{im} is the maximum inundation depth. For the numerical model presented here if the condition $h_{im}/D_F > 0.05$ is satisfied, the structure is categorised as being close to the shoreline. Thus, the equation that is applicable for an average estimation of force is

$$3. \quad F = 1.8 \rho g h_{im} B$$

where B is the breadth of the structure. As an appropriate safety factor, Fujima *et al.* (2009) suggests the coefficient of 1.8 in Equation 3 be increased to 3.3. These equations have resulted from small-scale physical experiments for bores propagating over a dry bed.

Robertson *et al.* (2011) categorised the main research on wave impact loads on structures into the following three areas: (a) work related with storm wave impact on offshore platforms that are presently the most commonly studied; (b) combining experimental and numerical research in order to develop design formulae for associated loads on structures; and (c) research on the forces and associated structural response resulting from tsunami bores impacting on land-based structures. This last research area is the least studied. Consequently, there is a need for better understanding of tsunami bore impact on onshore coastal structures due to the relatively limited available research. Following the Indian Ocean tsunami in 2004 and the Tohoku tsunami in 2011, extensive study has been conducted in order to improve the understanding of tsunami impact loads and improve design guidelines. However, research has mainly focused on experimental investigations and has been mostly confined to small-scale models; numerical models have been limited due to the high computational demand (Como and

Mahmoud, 2013). The research presented herein is intended to address the aforementioned issues by providing more detailed information on tsunami-induced bore and structure interactions by way of numerical models. The numerical study is conducted using two software packages: (a) the fluid is simulated using the smoothed particle hydrodynamics (SPH) software DualSPHysics; DualSPHysics is a free open-source SPH code released online (<http://www.dual.sphysics.org>), (b) the structural response is simulated using the commercial software Abaqus. The ultimate goal of the research is to enhance present understanding of tsunami wave–structure interaction with a view to improve current design provisions.

SPH is a meshless Lagrangian technique which is ideal for simulating highly non-linear free-surface phenomena such as tsunami waves. By solving the three-dimensional (3D) Navier–Stokes equations, the SPH numerical modelling technique offers the potential for improved definition of wave characteristics and associated pressures on impacted structures. The capabilities of SPH to model wave–structure interaction for coastal engineering problems were presented by Altomare *et al.* (2015), Barreiro *et al.* (2013) and Dalrymple *et al.* (2009). Previous studies by the authors have demonstrated the applicability of SPH in quantifying tsunami wave characteristics within acceptable levels of accuracy when impacting vertical onshore structures (Cunningham *et al.*, 2014). In addition to the laboratory experiments on tsunami wave impact on structures near shore, St-Germain *et al.* (2014) also conducted the numerical modelling using SPH on the basis of analogies between tsunami bores and dam-break waves. Furthermore, the SPH technique has also been used to model other violent wave behaviour such as storm wave impact on vertical walls near the shore (Altomare *et al.*, 2015). More details about DualSPHysics program can be found in Crespo *et al.* (2011, 2013, 2015) and Gómez-Gesteira *et al.* (2012a, 2012b). Although Canelas *et al.* (2013) have coupled SPH to the discrete-element method for fluid–structure interaction, none of the aforementioned studies have investigated the response of a deformable structure due to the hydrodynamics of a tsunami-like wave. This paper addresses that gap building on the work of Cunningham *et al.* (2014).

This paper is structured as follows: first, an overview of the two stages of numerical modelling is given. This is followed by detailed description of the SPH modelling, including model geometry and associated convergence study. Subsequent to this, the results from the SPH models – including the pressure–time histories that will be applied in the finite-element (FE) model – will be described. Finally, the response of the impacted structures by way of FE analysis will be presented and discussed.

2. Methodology

The numerical modelling conducted herein consists of two stages, the first being the simulation of tsunami-like waves

followed by the modelling of the response of the wave-impacted structure. The numerical simulations of the tsunami-like wave were conducted using DualSPHysics version 3.0 and were intended to predict tsunami-like bore pressure distributions on the surface of impacted structures. The output obtained from SPH numerical modelling, in the form of wave pressure distribution histories, are then used as an input in the second-stage numerical modelling using the commercially available FE software Abaqus/CAE 6.10 to define the structural response of an idealised residential structure. DualSPHysics can perform SPH computational modelling on both central processing units and graphics processing units (GPUs). Hardware acceleration provided by a GPU was performed to take advantage of speed ups of up to two orders of magnitude. In addition to the previous studies undertaken by the authors, further validation of the SPH model was conducted by remodelling the dam-break simulation done by Kleefsman *et al.* (2005).

For simulation purposes, throughout this work the tsunami wave is idealised as a solitary wave. This is an idealisation taking advantage of the well-defined and reproducible characteristics of a solitary wave. For discussion on the merits of this approach, the reader is referred to McCabe *et al.* (2014). The variation of solitary wave height, H , was non-dimensionalised using the offshore water depth h_0 . Three different cases with different solitary wave heights were performed: $H/h_0 = 0.1$, 0.3 and 0.5 . Other parameters in the modelling were kept constant. The H/h_0 values were expected to give associated bores with specific heights and velocities that are the two main factors in the determination of bore impact pressure on onshore structures. The numerical pressures output from each numerical case will be compared with the pressures determined from existing empirical equations that are based on physical experiments for bores propagating over a dry bed.

The tsunami-like wave simulation presented herein uses a small-scale domain. To implement the numerical modelling results in a real condition, normal similarity Froude scaling laws are applied. For practical purposes, the size of the idealised structure in the numerical SPH model ($1\text{ m} \times 1\text{ m} \times 1\text{ m}$) was scaled up by a factor of 3; this value was chosen to resemble the feasible size of a simple single storey residential shelter made of timber. In line with Froude scaling laws, the SPH pressure output was also scaled up by a factor of 3. The structure's dimensions and the pressure loads after being scaled are then used in the FE model. These types of structures were chosen as the focus of the FE modelling because they are fairly typical of the coastal buildings widely damaged by the tsunamis of 2004 and 2011 (Como and Mahmoud, 2013; Suppasri *et al.*, 2013). The dimensions and material properties of the representative timber structure's members in the FE model follow the provisions set out in the International Residential Code (ICC, 2009). More details of SPH and FE modelling are explained in the following sections.

3. Description of the SPH models

The SPH method is based on integral interpolants. Following Barreiro *et al.* (2013), the fundamental principle is to approximate any function $A(\mathbf{r})$ by

$$4. \quad \langle A(\mathbf{r}) \rangle = \int_{\Omega} A(\mathbf{r}') W(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}'$$

where $\mathbf{r}$ is the position, W is the weighting function or kernel, h is the smoothing length that controls the radius of influence of domain of Ω and $\langle \dots \rangle$ denotes an approximation. Equation 4, in discrete notation, leads to an approximation of the function at a particle of interest (interpolant point) a

$$5. \quad A_a = \sum_b A_b \frac{m_b}{\rho_b} W_{ab}$$

where the subscript refers to each particle, m is the mass, and ρ is density and the summation is performed over all the particles b within the region of compact support of the kernel function, as illustrated by Figure 1.

The SPH model relies on the selection of the weighting functions that must satisfy conditions such as positivity, compact support and normalisation (Monaghan, 1992). When evaluated for the interaction between two particles a and b , the weighting function, W_{ab} , depends on the smoothing length, h , and is normally expressed in terms of the non-dimensional distance between the particles given by $q = r_{ab}/h$, where r_{ab} is the distance between particles a and b ($r_{ab} = |\mathbf{r}_a - \mathbf{r}_b|$). The parameter h controls the size of the area surrounding particle a where the contribution of any other particles inside the area cannot be neglected. In the SPH simulations presented in this paper, the fifth-order Wendland kernel is used. More information on the kernels and the formulation available in DualSPHysics can be found in Crespo *et al.* (2015).

As in many weakly compressible SPH (WCSPH) simulations, the kernel approximation and its gradient are not corrected to ensure reproducibility of the constant and linear functions.

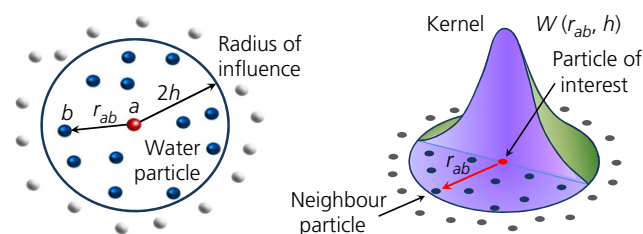


Figure 1. Schematic diagram of the SPH smoothing kernel

Although kernel correction is attractive mathematically, and in some SPH formulations essential (notably the incompressible SPH simulations where conservation is discarded in favour of increased accuracy, Lind *et al.*, 2012), it is not essential for WCSPH simulations since the SPH equations conserve the fundamental quantities of mass and momentum (Violeau and Rogers, 2016). Indeed, enforcing reproducibility can make WCSPH unstable while Altomare *et al.* (2015), Barreiro *et al.* (2013), Farahani *et al.* (2014) and St-Germain *et al.* (2014) have shown that WCSPH with no correction gives satisfactory results for nearshore wave processes.

The SPH formalism is used to discretise the governing equations expressing conservation of mass and momentum.

Conservation of mass

In WCSPH, the conservation of mass equation, or continuity, is solved in Lagrangian form as

$$6. \quad \frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v}$$

where $\mathbf{v}$ is the velocity, t is the time and ρ is the density. The changes in the fluid particle (FP) density are determined by solving the conservation of mass or continuity equation in SPH form

$$7. \quad \frac{d\rho_a}{dt} = \sum_b m_b \mathbf{v}_{ab} \cdot \nabla_a W_{ab}$$

where $\mathbf{v}_{ab} = \mathbf{v}_a - \mathbf{v}_b$ and $\nabla_a W_{ab}$ denotes the derivative of the smoothing kernel W_{ab} with respect to the coordinates of particle a . The mass of each SPH particle is kept constant.

Conservation of momentum

In Lagrangian form, the momentum equation in continuous form is

$$8. \quad \frac{d\mathbf{v}}{dt} = -\frac{1}{\rho} \nabla P + \mathbf{g} + \mathbf{\Gamma}$$

where P is the pressure, $\mathbf{g} = (0, 0, -9.81) \text{ ms}^{-2}$ is the gravitational acceleration and $\mathbf{\Gamma}$ is the dissipative term. There are several ways to solve the dissipative terms; however, the most widely used due to its simplicity is the artificial viscosity proposed by Monaghan (1992) and is used herein.

There are numerous forms of an SPH gradient that are chosen according to the physics or numerical properties. With the classical formulation, the pressure gradient in SPH notation uses

a symmetric form of the gradient to conserve momentum

$$9. \quad -\frac{1}{\rho} \nabla P = -\sum_b m_b \left(\frac{P_a}{\rho_a^2} + \frac{P_b}{\rho_b^2} \right) \nabla_a W_{ab}$$

where P_b and ρ_b are pressure and density of particle b . The artificial viscosity can be included in Equation 9 by adding the viscosity term Π_{ab} inside the bracket. Thus, the momentum conservation equation in SPH will be

$$10. \quad \frac{d\mathbf{v}_a}{dt} = -\sum_b m_b \left(\frac{P_a}{\rho_a^2} + \frac{P_b}{\rho_b^2} + \Pi_{ab} \right) \nabla_a W_{ab} + \mathbf{g}$$

The artificial viscosity depends on the relative position and motion of the computed particles

$$11. \quad \Pi_{ab} = \begin{cases} \frac{-\alpha \bar{c}_{ab} \mu_{ab}}{\rho_{ab}}, & \mathbf{v}_{ab} \mathbf{r}_{ab} < 0 \\ 0, & \mathbf{v}_{ab} \mathbf{r}_{ab} > 0 \end{cases}$$

where $\mathbf{v}_{ab} = \mathbf{v}_a - \mathbf{v}_b$, $\mu_{ab} = h \mathbf{v}_{ab} \mathbf{r}_{ab} / (r_{ab}^2 + \eta^2)$, $c_{ab} = 0.5(c_a + c_b)$ is the mean value of the speed of sound, $\eta^2 = 0.01 h^2$, and α is a parameter whose value ranges from 0.01 to 0.5, that should be adjusted according to the configuration of the problem.

In the simulations presented herein, a 3D SPH numerical water tank is used for simulating nearshore and onshore areas. In DualSPHysics, the implemented boundary conditions include the dynamic boundary conditions (DBC) for solid walls (Crespo *et al.*, 2007) and the periodic boundary conditions for lateral open boundaries (Gómez-Gesteira *et al.*, 2012a, 2012b). To represent a solid wall, the dynamic SPH boundary was used for the numerical model. The boundary particles and the FPs in the DBCs satisfy the same equations, but are not allowed to move (the movement is equal to zero) in any direction except where externally imposed such as a flap or a piston in a wavemaker. In the DualSPHysics code, the initial geometry, configuration and case parameters are defined in the extensible markup language (XML) input file and this also includes the size of particles. In the following subsections, first the particle size required to capture the pressure due to the bore impact is identified and second, this information is then used to model the tsunami bore impacting the structure.

3.1 Choosing the particle size (d_p): convergence study for bore impact

The choice of water particle size or the initial spacing of particle (d_p) is crucial in SPH modelling. The particle size

influences the overall simulation, including the behaviour of waves, the accuracy of pressure prediction and the time of simulation. The particle size was determined using a numerical convergence study by remodelling the dam-break impact on a box case in Kleefsman *et al.* (2005) as suggested by the SPH European Research Interest Community for relevant model validation. The use of the dam-break case for validation purposes is based on the assumption that the bore characteristics of the tsunami are closely represented by a broken dam-break wavefront (Chanson, 2006). This case has been used previously for validation to demonstrate convergence (Crespo *et al.*, 2011); here the maximum particle size required to capture the impact pressures is identified.

The fluid flow properties in the convergence study were the impact pressure and the water level. Many parameters influence the pressure and the water level, but in this case their values were kept constant for all models and only the particle size (d_p) was modified following previous work by the authors (Cunningham *et al.*, 2014). The only execution parameter recently added in this model is ‘ δ -SPH’ where the value was equal to 0.1. The δ -SPH is one of some improvements in DualSPHysics version 3 that was used for all SPH models in this paper. The δ -SPH scheme virtually eliminates unphysical pressure fluctuations and thereby, enhances the pressure prediction of an SPH model involving violent waves.

A convergence study is needed to determine the largest particle size and hence, the shortest run time that can be used in a simulation to capture hydrodynamics. The shortest dimension in the associated dam-break model that was used as a reference was the height of the impact-target structure in the form of a box equal to 0.161 m (Figure 2(a)). The height of the box is denoted herein by L . The convergence study for the pressure predicted at P1 is depicted in Figure 2(b) where three different values of d_p are compared. It can be seen that the d_p value equal to $L/20$ gives the closest prediction to the experiment results. Using d_p values smaller than $L/20$ shows little improvement and is too computationally expensive to be implemented. The GPU specification and run times are given in Table 1. A list of key constants and execution parameters in the XML input file used in this study are given in Table 2. The prediction of the smaller d_p values is not presented here. Following the results of the convergence study, the diameter of particle (d_p) representing fluids and solid parts in the model domain was set equal to 0.05 m or $1/20$ of the still water level (h_0) or the height of structure (D) that was, in this case, set to be 1 m; this led to 639 698 particles. Other SPH simulations that can be considered as convergence studies were the modelling of experimental works by Linton *et al.* (2013) and Zhang (2009) that can be found in previous work by the present authors (Cunningham *et al.*, 2014).

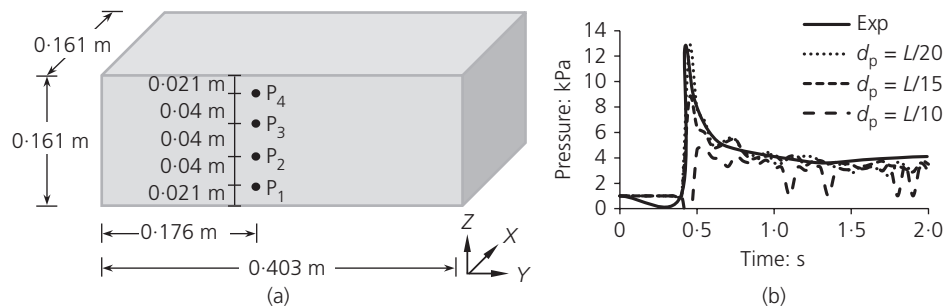


Figure 2. Dam-break case for SPH validation: (a) box dimension and probes position and (b) SPH pressure at probe P1

GPU specifications		SPH model details (DualSPHysics v3.0)	
GPU type	Tesla M2050	Particle size	0.05 m ($L/d_p = 1/20$)
Memory global	2687 Mb	Total particle	639 698
Number of cores	488	Time out	0.025
Clock rate	1.15 GHz	Total run time	36 min

Table 1. GPU specification used in the modelling and the run times

Constants		Execution parameters	
Particle size d_p : m	0.05 m	Step algorithm	2 (symplectic)
Lattice number	Bound = 2, fluid = 1	Kernel	2 (Wendland)
Courant-Friedrichs-Lewy number	0.2	Visco treatment	1 (artificial)
Coefficient of sound	10	Viscosity	0.1
Epsilon value	0.25	Shepard step	30

Table 2. Key XML parameters for input into DualSPHysics

3.2 Description of geometry of the SPH model

Having established the particle size required to capture the pressure history on the face of an obstacle similar to those being studied, the focus of the paper now shifts to simulating the input of a tsunami-type bore on a structure near the shoreline.

The geometry of the SPH numerical domain is shown in Figure 3 where a notation of D is used in the model to represent its characteristic length of 1 m. The size of model components or lengths in the geometry is given as a multiple of D , for example, the size of structure and the depth of still water are D , the distance of the structure to the shoreline and to the rear boundary wall are $2D$, and the width of water tank and

the height of paddle wavemaker are $3D$. This was intended to non-dimensionalise the analysis results. Figure 3 indicates the direction from where the pressure distribution can be viewed and this is the viewpoint (shown by the eye symbol) used for the snapshots that are shown later.

The dimensions of the numerical water tank here are 15 m long, 3 m wide and 5 m high. The numerical domain consists of a flat offshore region containing water particles, a sloping sea bed and onshore region where the coastal structure is placed. In a 3D model, the boundary line for the sloping bed (before shoreline) could be the location where water particle penetration occurs. To prevent an occurrence of excessive FPs out of the boundary, a double layer of particles is used for the

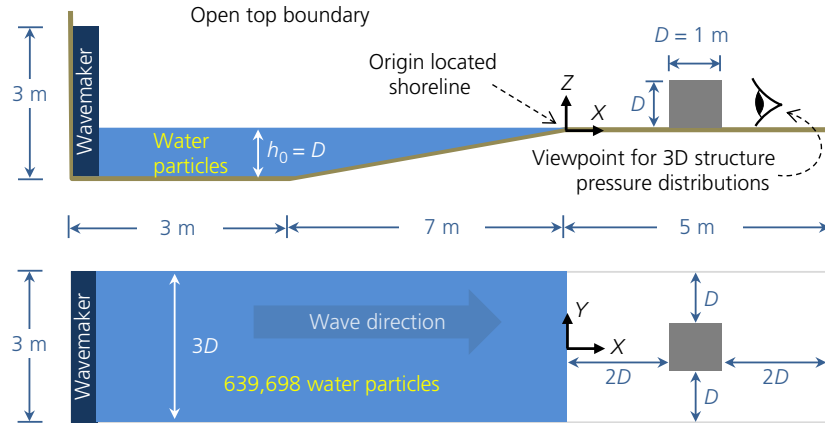


Figure 3. Side and plan view of 3D model with simple structure (not to scale)

boundary line by setting the ‘lattice number’ equal to two. Excessive out-of-boundary FPs could reduce the amount of FPs inside the tank and could affect the wave propagation (reduce the water surface elevation), especially when it undergoes a long propagation. For all models in this study, the percentage of the out of boundary FPs was limited to <1% of the total amount of FPs involved.

The parameter varied in the simulations is the wave height based on a ratio between targeted wave height and still water level (H/h_0). Three different values of H/h_0 were used: 0.1, 0.3 and 0.5, these provide the respective wave height 10, 30 and 50 cm for h_0 equal to 1 m. Those wave heights resulted in bores when reaching onshore and subsequently, impacted the onshore structure. A rectangular structure was chosen as representative of a coastal structure and hence, the impact target. The dimensions of the rectangular structure are $1\text{ m} \times 1\text{ m} \times 1\text{ m}$. The incoming wave was expected to hit the vertical sides of the cube that were perpendicular to the incoming wave direction. The distance of the vertical side of the cube was set at $2D$ (2 m) from the shoreline.

The origin for the numerical models lies within the model domain and is located at the shoreline. Since DualSPHysics is a single precision code, the position of zero axis can influence the accuracy of measurements of quantities such as pressure. In other words, a model with the zero axis located somewhere inside the domain performs better than when its zero-axis position is situated at the end of or much further from the model domain (Longshaw and Rogers, 2015). Another advantage of the zero-axis position inside the domain at the shoreline in this model is the ease of modifying parts of the model at both ends of the water tank. For example, when it is necessary to adjust the paddle distances from the shoreline or change the positions

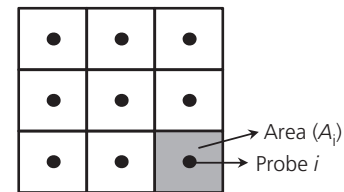


Figure 4. Arrangement of probes on the face of the obstacle

of the target structure (together with the measuring probes) at the opposite end of the water tank, it can be done without changing large parts of the XML input file.

4. Computing SPH pressures and forces on the structure

The DualSPHysics software provides output in the form of pressure–time history measured by numerical measuring probes. This pressure value can be used to estimate the force. As illustrated in Figure 4, the force (F_i) at a certain point in a 3D model can be determined by multiplying the pressure (P_i) with the area (A_i) of probe i (Equation 12). Hence, the total force (F) acting normal to a surface can be estimated by summing the force acting on the total number (n) of areas used to represent the object surface (Equation 13)

$$12. \quad F_i = P_i A_i$$

$$13. \quad F = \sum F_i = \sum_i^n (P_i A_i)$$

Numerical measuring probes were placed at several locations inside the model domain. The pressure probes were evenly distributed on all surfaces of the target structures shown in Section 6. The spacing of pressure probes on the structure's surface facing the incoming waves is 0.1 m or twice the diameter of particles as the interpolation region around each probe (this is approximately twice the smoothing length or particle size in radius) is sufficient to capture the peak pressure. For the vertically arranged measuring pressure probes, the probes start at a height of 0.05 m from the bed. Probes for measuring wave velocity and wave height were placed along the longitudinal axis of the tank (Figure 5).

5. SPH modelling results and discussion

5.1 Water surface elevation

Previous validation for the SPH water surface elevation generated by solitary wave was presented by Cunningham *et al.*

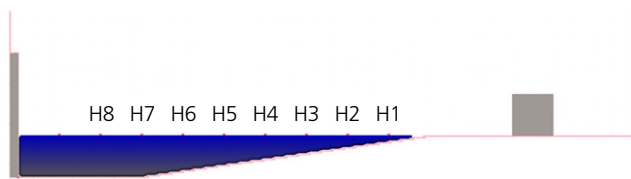


Figure 5. Layout of water surface elevation probes H1–H7

(2014). Here, the authors present the results for different wave height to water depth ratios. Figure 5 shows the probe positions for measuring water surface elevation regarding the propagation of the solitary wave for cases with different H/h_0 . The water surface elevations were measured at certain probes along the tank. H1 is placed 1 m from the shoreline and followed by H2 through H7 placed at a constant 1 m spacing apart. The properties of the offshore solitary waves and onshore bores can be seen from Figures 6 and 7, and also from Table 3. Figure 6 depicts the SPH simulation of the solitary wave propagation offshore for the case with $H/h_0 = 0.5$. From the four snapshots given in Figures 6(a)–6(d), the changes in the solitary wave profile/shape can be observed. Near the shoreline at probe H1, the solitary wave started to break and its elevation decreased as illustrated in more detail in Figure 7.

Figure 7 shows the comparison of solitary wave elevations based on different H/h_0 during propagation towards the shoreline. The surface elevations were measured at four different probes H7, H5, H3 and H1. The wave elevations were measured from the surface of water at rest. From H7 until H1, it can be seen that generally the solitary wave elevation slowly decreases. The maximum elevations of the solitary wave for the case with $H/h_0 = 0.5$ is almost 100% higher than the case with $H/h_0 = 0.3$, and almost three times higher than the case with $H/h_0 = 0.1$. The differences in wave elevation correspond

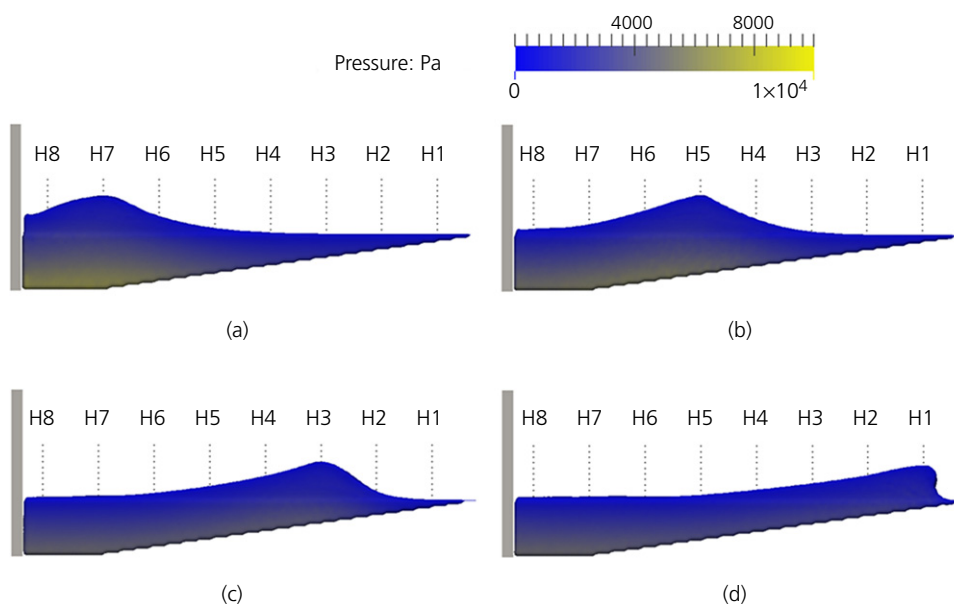


Figure 6. SPH simulation of solitary wave height and pressure during propagation from probe H7 through probes H1 for case with $H/h_0 = 0.5$: (a) maximum wave elevation at probe H7;

(b) maximum wave elevation at probe H5; (c) maximum wave elevation at probe H3; and (d) maximum wave elevation at probe H1

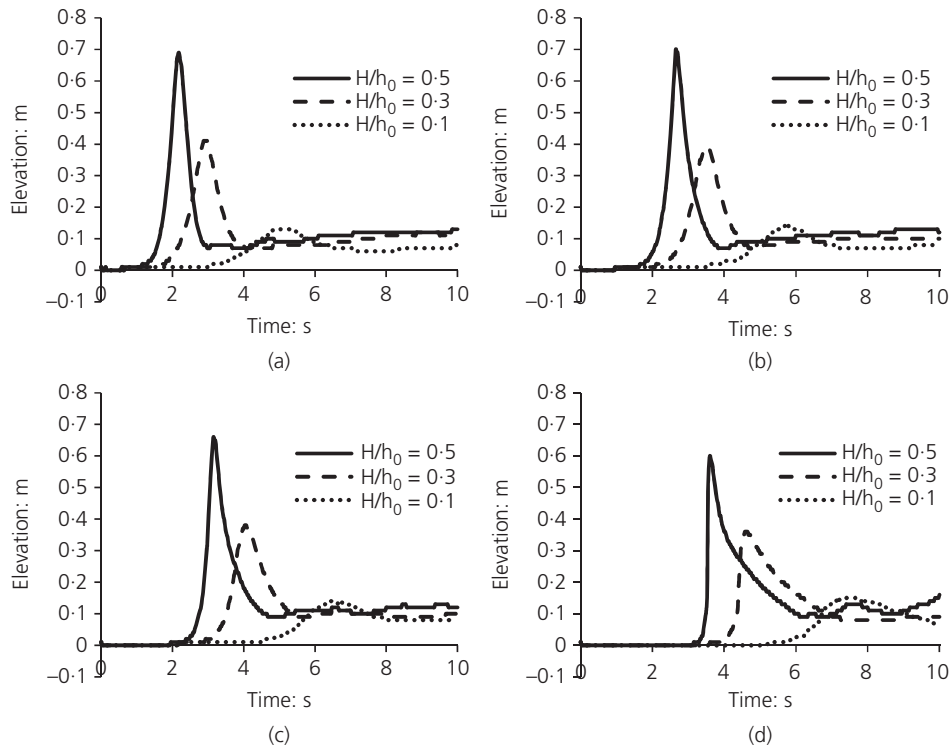


Figure 7. Comparison of solitary wave elevations for different H/h_0 measured at probe H7, H5, H3 and H1: (a) probe H7; (b) probe H5; (c) probe H3; (d) probe H1

H/h_0	Offshore solitary waves			Onshore bores	
	Maximum elevation: m	Maximum velocity: m/s	Wave length: m	Maximum height: m	Maximum velocity: m/s
0.5	0.7	3.94	5.0	0.32	5.65
0.3	0.4	2.13	6.5	0.20	4.05
0.1	0.15	0.68	7.7	0.11	1.19

Table 3. Solitary wave and bore properties

to the differences of the associated bore impact forces on the structure. It can also be seen that there are time differences regarding the maximum wave elevations when they reach the probes.

5.2 Bore height and velocity

The onshore bore heights and velocities were measured by probes placed in front of the structure. The values of these properties are important in determining the exerted pressure on the surface of structure. The snapshots of the bores,

including velocities, can be seen in Figure 8 and the variation of bore heights are depicted in Figure 9 for simulation cases with $H/h_0=0.3$ and 0.5 . Table 3 shows the offshore water surface elevation and wave length of solitary waves, and the corresponding onshore bore characteristics (height and velocity). It can be observed that offshore, the maximum velocity increases proportionally with the design height of the solitary wave, but is inversely proportional to the wave length of the wave. The maximum solitary wave velocities offshore are also proportional to the bore height and velocity onshore.

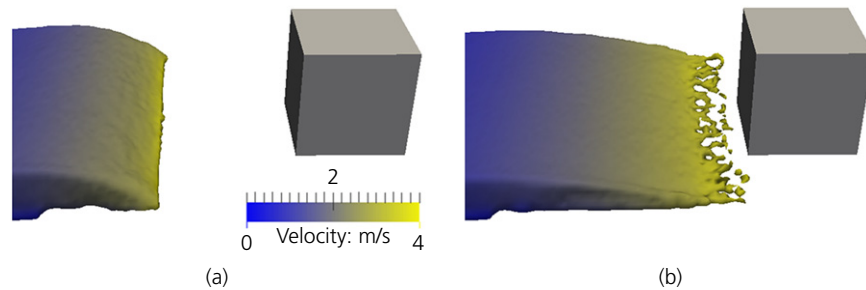


Figure 8. SPH simulation of bore velocity for the case with $H/h_0 = 0.5$: (a) near the shoreline and (b) onshore

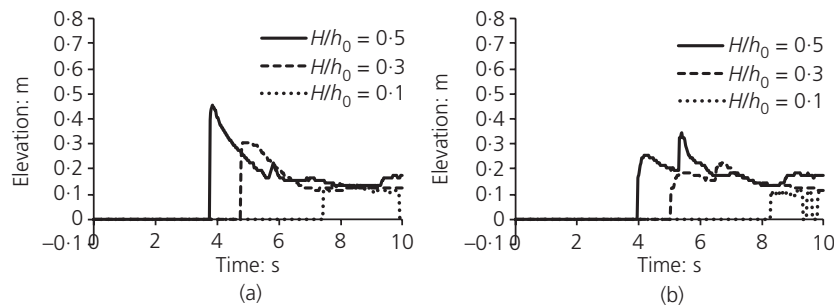


Figure 9. Bore height measured from still water level at (a) shoreline and (b) onshore

5.3 Wave pressure

Typical output from the 3D SPH simulation is shown in Figures 10 and 11 for the case with $H/h_0 = 0.5$. Figure 10(a) depicts the snapshot of a propagating solitary wave that is followed by a bore impact on the structure as shown by Figure 10(b). The peak pressure impact occurred at $t = 4.150$ s and the peak pressure took place at the lowest level of pressure probes, as indicated by the circle in Figure 10(a). In addition, Figure 11(b) shows the pressure distribution for the maximum bore run-up on the surface of the structure that occurred at $t = 4.325$ s, corresponding with Figure 10(b). Pressure distributions in Figure 11 were seen from the rear of the structure (see illustration indicating direction of view in Figure 3) by assuming the cube structure is visually transparent.

The normalised peak pressures ($P/\rho gh_0$) for the simulation with $H/h_0 = 0.1$, 0.3 and 0.5 were 0.748 , 18.165 and 83.287 , respectively. The bore pressures for the case with $H/h_0 = 0.1$ were comparatively small and resulted in a correspondingly small impact force. Hence, the comparison of the predicted impact pressure between the numerical and the empirical equation were made only for the case with $H/h_0 = 0.3$ and

0.5 . The numerical maximum pressure for the case with wave height equal to $H/h_0 = 0.3$ and 0.5 were $10\,508$ and $37\,371$ N/m², respectively. The associated pressures determined by Equation 2 were 1766 and 4520 N/m² over the bore height for the case with $H/h_0 = 0.3$ and 0.5 , respectively. In addition, the impact pressures predicted by Equation 3 were 3532 and 5651 N/m² over the bore height for the case with $H/h_0 = 0.3$ and 0.5 , respectively. However, these results must be viewed within the context that, in general, the design equations adopt a quasi-static approach to wave pressures, whereas in reality the peak pressure occurs over a very short time period and may be very localised.

6. FE modelling

The FE modelling was conducted to study the behaviour of a simple dwelling-type structure made of timber under tsunami-like wave bore impact loading. The response of the structures, especially the stress distribution on the main load-bearing components – that is, the vertical studs, will be compared with the behaviour resulting from the quasi-static pressure given by Equation 2.

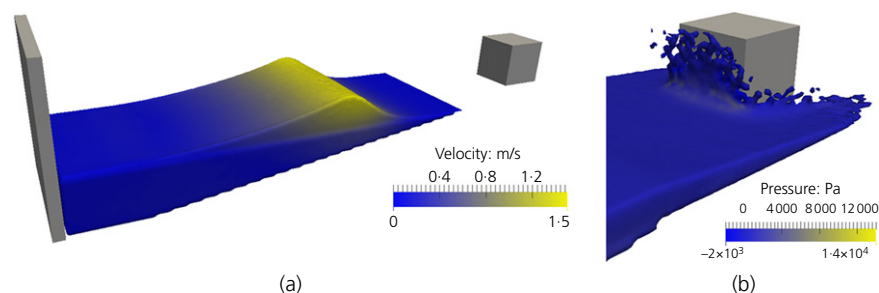


Figure 10. Oblique view of the 3D simulation for case $H/h_0 = 0.5$:
(a) solitary wave propagation at $t = 3.150$ s (related with
Figure 6(c)) and (b) bore impacting structure at $t = 4.325$ s

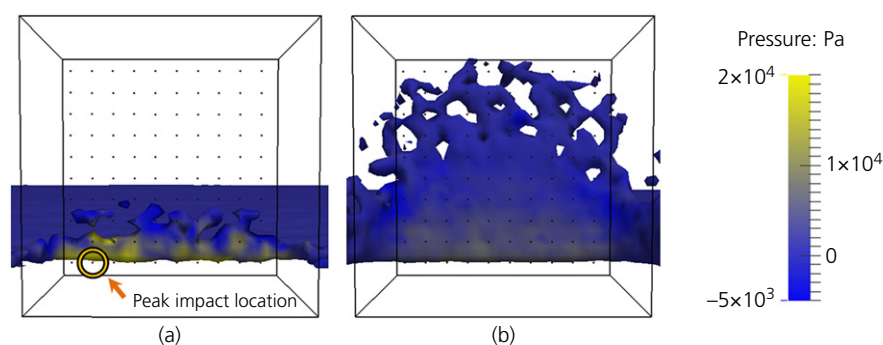


Figure 11. Pressure distribution on vertical surface at $H/h_0 = 0.5$:
(a) first impact at $t = 4.150$ s and (b) peak impact occurred at the
circled probe at $t = 4.325$ s

The dynamic analysis was performed using Abaqus Explicit (Abaqus, 2010). In this procedure, the equations of motion are integrated in time explicitly using central difference integration rules. The analysis procedure requires a small time increment and since it does not need to solve a global set of equations in each increment, the computational cost per increment is relatively small compared with the implicit method. The explicit method is well suited to modelling short duration/transient loads such as those associated with tsunami wave impacts.

6.1 Model properties

Figure 12 shows a definition sketch of the timber structure. As previously stated, the structure dimensions for the FE model are $3\text{ m} \times 3\text{ m} \times 3\text{ m}$. The target structure consists of four vertical walls and a flat roof all composed of sandwich timber panels. The sandwich panel is an arrangement of sheath–stud–sheath. The sheath is made of plywood with 13 mm thickness and has the following material properties: density = 750 kg/m^3 , Young's modulus = $7.7 \times 10^9\text{ N/m}^2$ and

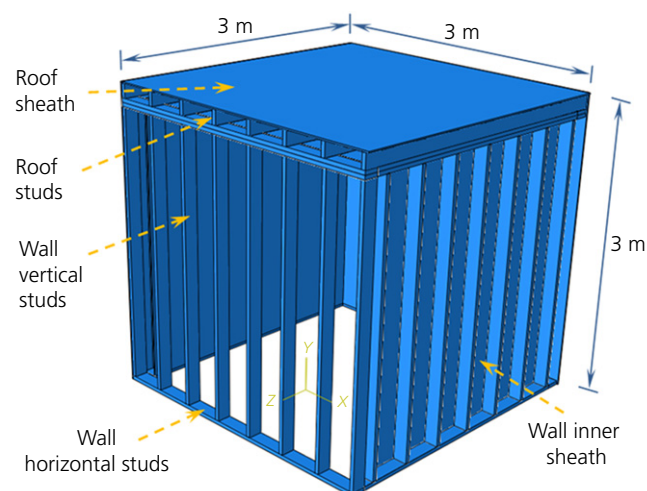


Figure 12. Timber structure general arrangement (wall sheaths removed for clarity, as shown)

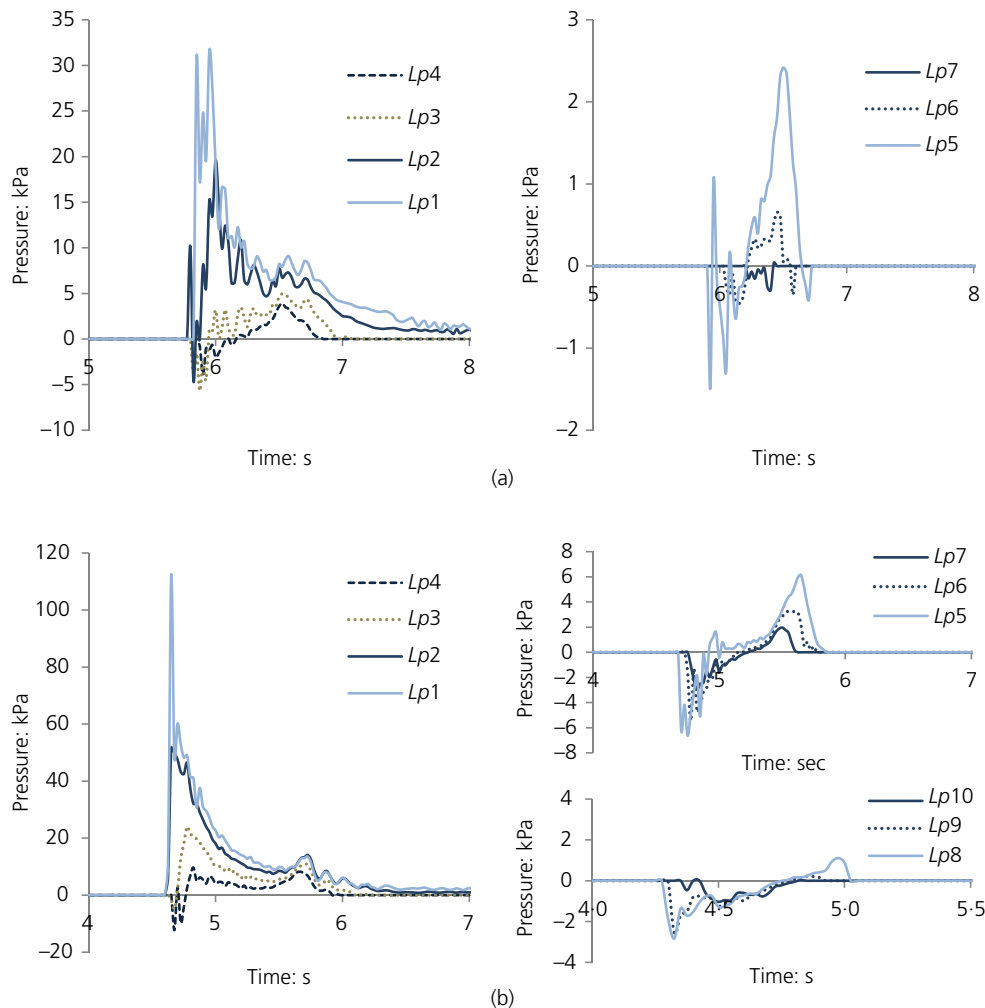


Figure 13. Applied pressure–time histories for the FE models:
(a) $H/h_0 = 0.3$ and (b) $H/h_0 = 0.5$

Poisson's ratio = 0.22. The studs are made of softwood timber (Douglas fir) with a cross-sectional size of 38 mm width $\times$ 140 mm depth, and installed with spacing centres of 420 mm. The studs' material properties are as follows: density = 530 kg/m³, Young's modulus = 7.0×10^9 N/m² and Poisson's ratio = 0.22. The timber structure was designed to be simply supported at the base – that is, no rotational fixity. The timber structure components were modelled using eight-node linear brick elements (type C3D8R) with reduced integration and hour-glass control.

6.2 Input loads

The applied pressure–time histories on the FE model structures were obtained from the DualSPHysics output. As mentioned earlier in Section 2, the size of structure in the FE model was scaled up by a factor of 3 from the structure in the

DualSPHysics model. Thus, following dimensional analysis, the magnitude of applied pressures on the surface of the FE models was also scaled up by three from the DualSPHysics output pressures. The pressure–time histories of applied loads can be seen in Figure 13, for both wave simulation cases with $H/h_0 = 0.3$ and 0.5. The output pressure from the wave simulation with $H/h_0 = 0.1$ was not included in the FE model analysis because the magnitude was relatively small. The FE structure responses were studied by applying two different pressure time histories (as shown in Figure 13) on the front surface of the structure. One of the key aspects in this method is the application of the same sizes for pressure probe spacing in the DualSPHysics model and the mesh element in the Abaqus model. Using the same particle and mesh size enables the output pressure values from the DualSPHysics 'MeasureTool' to be applied directly to the element in the

Abaqus model. The 'MeasureTool' provides pressure values over time in the form of tabular data for each SPH pressure probe. The pressure tabular data for each probe is transferred to the Abaqus element by using the 'load amplitude' and 'tabular' options in Abaqus. The simulations were then run using the dynamic explicit time stepping explained previously.

The pressures applied on the vertical surface of the structures were divided into ten layers. This number of layers was identical with the number of probes arranged vertically in the SPH model, denoted by L_p in Figure 13. The magnitude of pressure at each layer is the average pressure measured by pressure probes at the associated layer. The reason for applying average pressure history at each layer was based on time efficiency. Also, it was found that the pressure variation measured by SPH probes within the same layer is generally not significant. Figure 13(a) shows the applied loads presented from layer 1 to layer 7 (L_{p1} – L_{p7}). Loads at layer 8 to layer 10 (L_{p8} – L_{p10}) are zero so they were not included in the graph. In Figure 13(b), all ten layers were loaded. The loaded layers on the surface were related with the height of bore run-up on the surface of the wall.

The negative pressure values in Figure 13 reflect the application of the equation of state in DualSPHysics and a rebound or suction effect. DualSPHysics version 3.0 is still a single-phase model that cannot yet perform multiphase behaviour, including air and water at the same time. The next version of DualSPHysics (version 4.0) is being developed to be capable of performing such multi-phase modelling (Mokos *et al.*, 2015). Two peaks are shown in the pressure–time histories in Figure 13. The first peak occurred when the leading edge of the bore impacted the structure, and the second peak occurred when the splashing followed with the main flow. These two peak phenomena are typical for impact loads of this type and were also observed in the experiment by Fujima *et al.* (2009). Note that in the case of the immediate impact on the structure, no additional force is applied – for example, drag force. The hydrodynamic 'drag' forces indeed occur in real tsunami events (Yeh *et al.*, 2014) and these affect the structure when surrounded by a 'steady' fluid flow with relatively constant velocity (FEMA, 2011). However, in this simulation, the hydrodynamic drag was not so significant (Figure 14) due to the application of a single stroke solitary wave where the most dominant effect was the highly transient impulsive pressure immediately following the impact of the leading edge of the arriving water mass.

6.3 Analysis and results

The FE analysis was performed using the Abaqus dynamic explicit module with duration of simulation of 2.6 s and time increment automatically determined by the program. The

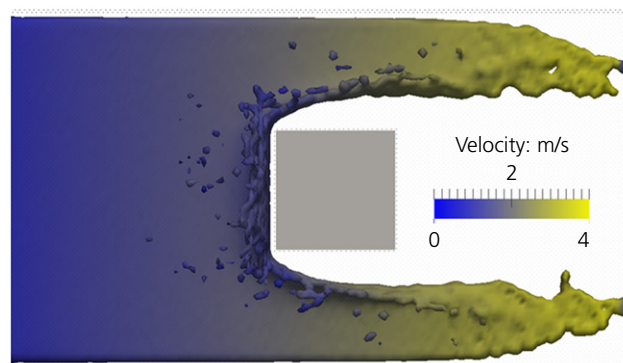


Figure 14. Plan view of SPH simulation showing no significant drag force on the sides of structure

duration of simulation was enough to capture the important segment of pressure–time histories, given in Figure 13, incorporating the peak impact and subsequent impact. All components of the structure were meshed with the finest mesh on the front surface. The size of the mesh for the front impacted surface was 0.05 m to correspond with the diameter of SPH FPs, a coarser mesh with twice the size of elements was used for the other sides of the structure. The response of the structures is shown in terms of the maximum principal stress that occurred on the members. On the front face of the structure, the main load carrying elements are the vertical and horizontal studs; these are shown in Figure 15 by removing the outer plywood cover sheath.

The results of the FE modelling can be seen in Figure 15 that has five figures showing the response of the structure under different loading conditions. Figures 15(a) and 15(b) shows the response of the structures loaded by the SPH pressure–time history resulting from a normalised wave height $H/h_0 = 0.3$ and 0.5, respectively. Figures 15(c) and 15(d), on the other hand, show the response of the structure under a quasi-static pressure determined using the semi-empirical approach given by Equation 2 using the equivalent wave properties. The pressures related with Figures 15(c) and 15(d) were calculated based on a bore height of 0.60 and 0.96 m, respectively, where these bore heights were taken from the SPH model simulation (Table 3) and scaled up by a factor of 3. The application of these bore heights in Equation 2 was intended to achieve a straightforward comparison of the structure's response based on numerical and semi-empirical approaches.

Figures 15(a) and 15(b) show that the maximum principal stresses occurred at the tension side, below the mid-span of the vertical studs. The stresses shown in Figures 15(a) and 15(b) are the maximum stresses that occurred at any time in the response

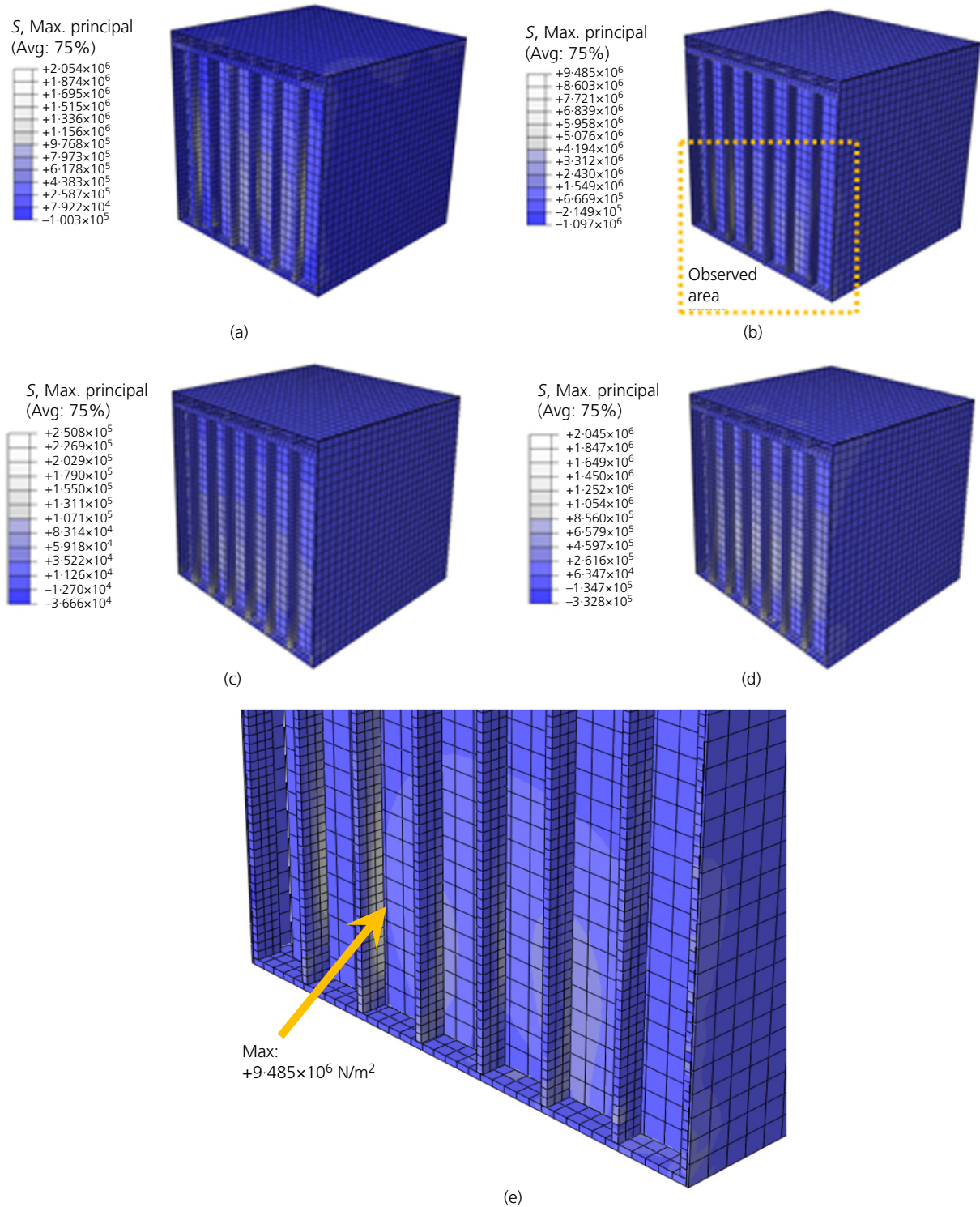


Figure 15. FE dynamic response due to SPH pressure–time history (a, b); FE static response due to quasi-static pressure from Equation 2 (c, d); location of maximum stresses (e); (all stresses shown in N/m²)

to the bore impact and are not necessarily simultaneous with the maximum impact pressure. In the case of Figure 15(b), the stresses are approaching the flexural capacity for this type of timber. This general type of flexural failure is in line with experimental results reported by Lacroix and Doudak (2014) whereby the vertical studs on timber panels failed due to the out-of-plane high strain rate induced by blast loading; in this case, the tsunami impact load could also be categorised as a high strain rate event as indicated by the pressure–time histories in Figure 13. For comparable scale structures, similar failure patterns of timber studs in flexure due to tsunami-like wave impact was reported experimentally by Linton *et al.* (2013). It can also be seen that under higher loads (Figure 15(b)), the stress concentration tends to develop closer to the lower end of the studs in comparison with Figure 15(a). In this case, a shear-type failure of the stud would be expected. Figures 15(c) and 15(d) show the response of the structure under the static loads resulting from Equation 2. It can be seen that the maximum stresses concentrated at the bottom end of the vertical studs. This response is due to load being applied on the lower part of the structure over the height of the bore. In both Figures 15(c) and 15(d), it should be noted that the predicted quasi-static pressures resulted in timber stresses at least four times less than those observed in the dynamic response using the SPH-derived pressure–time histories. In real terms, these values represent the difference between the focused structure remaining intact or not.

7. Conclusions

In this paper, the interactions between a tsunami bore and an idealised timber structure were modelled using the SPH-based software DualSPHysics coupled with FE analysis using Abaqus. The SPH models show the ability of this approach to provide impact pressure distributions in great detail, both spatially and temporally. The results from the SPH models were utilised in the FE model to predict the response of the structure under transient bore impact loads. The numerical pressure predictions were also compared with pressure predictions from semi-empirical approaches used in design codes. The structure response results highlight the potential for non-conservatism in the empirical approaches. The techniques outlined in this paper can allow engineers to gain a better insight into tsunami wave–structure interaction and thus lead to resilience optimisation of structures vulnerable to these type of events. Further development is needed for more complex tsunami-like bore and structure interactions, including the effect of aeration by way of multi-phase SPH wave modelling.

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